

OPDI Papers

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Methodology studies behind the [Open Performance Data Initiative](#), covering how OPDI turns raw ADS-B state vectors into flight tracks.

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[Turning ADS-B state vectors into flight tracks — version 1](#) · [PDF](#)

The eight-rule benchmark that selected the shipped segmentation. Scores eight candidate rules for grouping state vectors into tracks against three days of Network Manager ground truth, finds that the rule behind every `track_id` OPDI has published to date recovers only 50.15% of flights one-to-one on 2025, and that grouping on the airframe alone instead of (`icao24`, `callsign`) raises that to 90.80%. That rule now ships as `segmentation.method = "standard"`.

[A new rule builds OPDI's flight tracks — version 2](#) · [PDF](#)

The release note: the real pipeline reproduces V1's harness result. Runs `legacy`, `airframe_only` and `standard` through the full production chain — steps 01 through 03, not a harness over an already-cleaned table — on two independent days a year apart, and finds the pipeline's gain for `standard` over `legacy` agrees with V1's harness figure to within 1.08 points on all four comparisons (airborne and gate-to-gate, 2025 and 2024). States what changes for a consumer: `track_id` stops carrying a `_{year}_{month}` suffix, no `track_id` published before the release date joins against data built after it, and nothing in `osn_tracks` marks which rule produced a row.